

[原创]AD7705/AD7706的使用

今天有点紧急，要处理一下AD7705刷新速率问题
我先介绍一下AD7705/AD7706，就是它了↓



ANALOG
DEVICES

3 V/5 V, 1 mW, 2-/3-Channel,
16-Bit, Sigma-Delta ADCs

AD7705/AD7706

FEATURES

AD7705: 2 fully differential input channel ADCs
AD7706: 3 pseudo differential input channel ADCs
16 bits no missing codes
0.003% nonlinearity
Programmable gain front end: gains from 1 to 128
3-wire serial interface
SPI®, QSPI™, MICROWIRE™, and DSP-compatible
Schmitt-trigger input on SCLK
Ability to buffer the analog input
2.7 V to 3.3 V or 4.75 V to 5.25 V operation
Power dissipation 1 mW maximum @ 3 V
Standby current 8 μA maximum
16-lead PDIP, 16-lead SOIC, and 16-lead TSSOP packages

GENERAL DESCRIPTION

The AD7705/AD7706 are complete analog front ends for low frequency measurement applications. These 2-/3-channel devices can accept low level input signals directly from a transducer and produce serial digital output. The devices employ a Σ-Δ conversion technique to realize up to 16 bits of no missing codes performance. The selected input signal is applied to a proprietary, programmable-gain front end based around an analog modulator. The modulator output is processed by an on-chip Σ-Δ modulator and a digital filter. The digital filter output is processed by a serial interface. The serial interface is a 3-wire interface consisting of SCLK, CS, and DIN. The digital filter output is also available as a DOUT. The devices can accept bipolar or unipolar input signals. The devices can accept bipolar or unipolar input signals. The devices can accept bipolar or unipolar input signals.

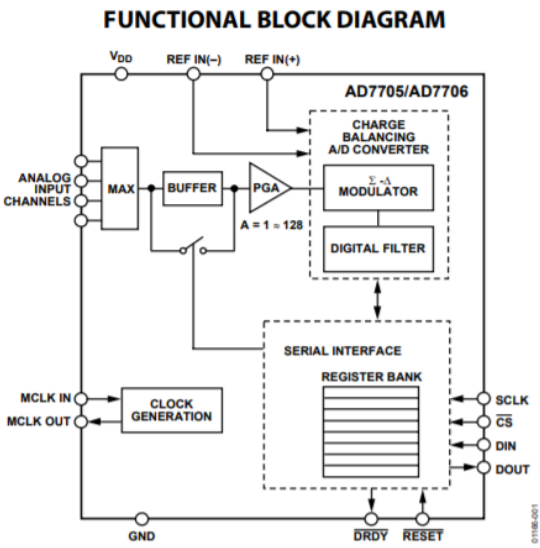


Figure 1.

The AD7705/AD7706 devices, with a 3 V supply and a 1.225 V reference, can handle unipolar input signal ranges of 0 mV to 10 mV through 0 V to 1.225 V. The devices can accept bipolar

AD7705两路全差分，AD7706三路伪差分

双路16位ADC模块
0.003%非线性,16位无丢码
500HZ更新速率 可编程增益



3.3V/5V电压兼容
全新原装进口AD7705
1mv/3ppmAD780BR电压基准

硬件设计：
说在前面：只为说明问题，没有别的意思。
先放一个TB的模块，蓝字我就不念了。大部分都是AD7705芯片本身的属性，但这里说3.3V和5V兼容，我们看看怎么兼容：

AD7705/AD7706			
Parameter	B Version ¹	Unit	Conditions/Comments
AIN Input Sampling Rate, f _s	Gain × f _{CLKIN} /64 f _{CLKIN} /8		For gains of 1 to 4 For gains of 8 to 128
Reference Input Range			
REF IN(+) – REF IN(–) Voltage	1/1.75	V min/V max	V _{DD} = 2.7 V to 3.3 V V _{REF} = 1.225 ± 1% for specified performance
REF IN(+) – REF IN(–) Voltage	1/3.5	V min/V max	V _{DD} = 4.75 V to 5.25 V V _{REF} = 2.5 ± 1% for specified performance
REF IN Input Sampling Rate, f _s	f _{CLKIN} /64		

芯片5V供电时,REF建议电压范围是1V~3.5V，推荐值2.5V
芯片3V供电时,REF建议电压范围是1V~1.75V，推荐值1.225V
模块上参考电压芯片是REF192-2.048，也就是2.048V，显然都不是推荐值，且这已经超出了3V供电的推荐范围，因此电路设计不规范。
(没别的意思，只是自己用网上的模块，又花时间整理，不如就详细点哈哈)
那ADI官方如何推荐？

1 Recommended reference voltage sources for the AD7705/AD7706
2 with a VDD of 5 V include the AD780, REF43, and REF192;
3 the recommended reference sources for the AD7705/AD7706
4 operated with a VDD of 3 V include the AD589 and AD1580.

5 It is generally recommended to decouple the output of
6 these references to reduce the noise level further.

第二个不规范

LOGIC INPUTS			
Input Current			
All Inputs, Except MCLK IN	±1	µA max	Typically ±20 nA
MCLK IN	±10	µA max	Typically ±2 µA
All Inputs, Except SCLK and MCLK IN			
Input Low Voltage, VINL	0.8	V max	VDD = 5 V
	0.4	V max	VDD = 3 V
Input High Voltage, VINH	2.0	V min	VDD = 3 V and 5 V
SCLK Only (Schmitt-Triggered Input)			VDD = 5 V nominal
VT+	1.4/3	V min/V max	
VT-	0.8/1.4	V min/V max	
VT+ - VT-	0.4/0.8	V min/V max	
SCLK Only (Schmitt-Triggered Input)			VDD = 3 V nominal
VT+	1/2	V min/V max	
VT-	0.4/1.1	V min/V max	
VT+ - VT-	0.375/0.8	V min/V max	
MCLK IN Only			VDD = 5 V nominal
Input Low Voltage, VINL	0.8	V max	
Input High Voltage, VINH	3.5	V min	
MCLK IN Only			VDD = 3 V nominal
Input Low Voltage, VINL	0.4	V max	
Input High Voltage, VINH	2.5	V min	
LOGIC OUTPUTS (Including MCLK OUT)			
Output Low Voltage, VOL	0.4	V max	ISINK = 800 µA, except for MCLK OUT; ¹³ VDD = 5 V
Output Low Voltage, VOL	0.4	V max	ISINK = 100 µA, except for MCLK OUT; ¹³ VDD = 3 V
Output High Voltage, VOH	4	V min	ISOURCE = 200 µA, except for MCLK OUT; ¹³ VDD = 5 V
Output High Voltage, VOH	VDD - 0.6	V min	ISOURCE = 100 µA, except for MCLK OUT; ¹³ VDD = 3 V
Floating State Leakage Current	±10	µA max	
Floating State Output Capacitance ¹⁴	9	pF typ	
Data Output Coding	Binary		Unipolar mode
	Offset binary		Bipolar mode

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红框：5V供电时，输出高电平最低是4V，你进3.3V系统，不串个限流电阻？
绿框：如果是晶振输入时钟，倒是无所谓。但是如果是外部芯片提供的时钟，那高电平最低要3.5V的，3.3V系统也存在使用风险的

软件设计：

AD7705本质上就是几大寄存器，但所有寄存器都要跟通信寄存器打声招呼，告诉通信寄存器，你打算干什么

1.通信寄存器000

Table 9. Communication Register

0/DRDY (0)	RS2 (0)	RS1 (0)	RS0 (0)	R/W (0)	STBY (0)	CH1 (0)	CH0 (0)
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Table 10. Communication Register Bit Description

Register	Description
0/DRDY	For a write operation to the communications register, a 0 must be written to this bit. If a 1 is written to this bit, the part does not clock subsequent bits into the register. It stays at this bit location until a 0 is written. Then, the next seven bits are loaded into the communication register. For a read operation, this bit provides the status of the DRDY flag, which is the same as the DRDY output pin.
RS2–RS0	Register Selection Bits. These bits are used to select which of the AD7705/AD7706 registers are being accessed during the serial interface communication.
R/W	Read/WRITE Select. This bit selects whether the next operation is a read or write operation. A 0 indicates a write cycle for the next operation to the selected register, and a 1 indicates a read operation from the selected register.
STBY	Standby. Writing 1 to this bit puts the part into standby or power-down mode. In this mode, the part consumes only 10 µA of power supply current. The part retains its calibration coefficients and control word information when in standby. Writing 0 to this bit places the parts in normal operating mode.
CH1, CH0	Channel Select. These two bits select a channel for conversion or for access to the calibration coefficients, as outlined in Table 12. Following a calibration on a channel, three pairs of calibration registers store the calibration coefficients. Table 12 (for the AD7705) and Table 13 (for the AD7706) show which channel combinations have independent calibration coefficients. With CH1 at Logic 1 and CH0 at Logic 0, the AD7705 looks at the AIN1(–) input internally shorted to itself, while the AD7706 looks at the COMMON input internally shorted to itself. This can be used as a test method to evaluate the noise performance of the parts with no external noise sources. In this mode, the AIN1(–)/COMMON input should be connected to an external voltage within the allowable common-mode range for the parts.

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写通信寄存器，将要操作的寄存器是哪个，读还是写，针对哪个通道？
然后你就可以开搞了

Table 11. Register Selection

RS2	RS1	RS0	Register	Register Size
0	0	0	Communication register	8 bits
0	0	1	Setup register	8 bits
0	1	0	Clock register	8 bits
0	1	1	Data register	16 bits
1	0	0	Test register	8 bits
1	0	1	No operation	
1	1	0	Offset register	24 bits
1	1	1	Gain register	24 bits

Table 12. Channel Selection for AD7705

CH1	CH0	AIN(+)	AIN(–)	Calibration Register Pair
0	0	AIN1(+)	AIN1(–)	Register Pair 0
0	1	AIN2(+)	AIN2(–)	Register Pair 1
1	0	AIN1(–)	AIN1(–)	Register Pair 0
1	1	AIN1(–)	AIN2(–)	Register Pair 2

Table 13. Channel Selection for AD7706

CH1	CH0	AIN	Reference	Calibration Register Pair
0	0	AIN1	COMMON	Register Pair 0
0	1	AIN2	COMMON	Register Pair 1
1	0	COMMON	COMMON	Register Pair 0
1	1	AIN3	COMMON	Register Pair 2

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2.设置寄存器001

Table 14. Setup Register

MD1 (0)	MD0 (0)	G2 (0)	G1 (0)	G0 (0)	B/U (0)	BUF (0)	FSYNC (1)
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Table 15. Setup Register Description

Register	Description
MD1, MD0	ADC Mode Bits. These bits select the operational mode of the ADC as outlined in Table 16.
G2 to G0	Gain Selection Bits. These bits select the gain setting for the on-chip PGA, as outlined in Table 17.
B/U	Bipolar/Unipolar Operation. A 0 in this bit selects bipolar operation; a 1 in this bit selects unipolar operation.
BUF	Buffer Control. With this bit at 0, the on-chip buffer on the analog input is shorted out. With the buffer shorted out, the current flowing in the V _{DD} line is reduced. When this bit is high, the on-chip buffer is in series with the analog input, allowing the input to handle higher source impedances.
FSYNC	Filter Synchronization. When this bit is high, the nodes of the digital filter, the filter control logic, the calibration control logic, and the analog modulator are held in a reset state. When this bit goes low, the modulator and filter start to process data, and a valid word is available in 3 × 1/output rate, that is, the settling time of the filter. This FSYNC bit does not affect the digital interface and does not reset the DRDY output if it is low.

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AD7705模式设置，主要是校准用的

AD7705的增益设置、极性输入设置、BUFFER控制、Filter 同步设置

Table 16. Operating Mode Options

MD1	MD0	Operating Mode
0	0	Normal Mode. In this mode, the device performs normal conversions.
0	1	Self-Calibration. This activates self-calibration on the channel selected by CH1 and CH0 of the communication register. This is a one-step calibration sequence. When the sequence is complete, the part returns to normal mode, with both MD1 and MD0 returning to 0. The DRDY output or bit goes high when calibration is initiated, and returns low when self-calibration is complete and a new valid word is available in the data register. The zero-scale calibration is performed at the selected gain on internally shorted (zeroed) inputs, and the full-scale calibration is performed at the selected gain on an internally generated V _{REF} /selected gain.
1	0	Zero-Scale System Calibration. This activates zero-scale system calibration on the channel selected by CH1 and CH0 of the communication register. Calibration is performed at the selected gain on the input voltage provided at the analog input during this calibration sequence. This input voltage should remain stable for the duration of the calibration. The DRDY output or bit goes high when calibration is initiated, and returns low when zero-scale calibration is complete and a new valid word is available in the data register. At the end of the calibration, the part returns to normal mode, with both MD1 and MD0 returning to 0.
1	1	Full-Scale System Calibration. This activates full-scale system calibration on the selected input channel. Calibration is performed at the selected gain on the input voltage provided at the analog input during this calibration sequence. This input voltage should remain stable for the duration of the calibration. The DRDY output or bit goes high when calibration is initiated, and returns low when full-scale calibration is complete and a new valid word is available in the data register. At the end of the calibration, the part returns to normal mode, with both MD1 and MD0 returning to 0.

Table 17. Gain Selection

G2	G1	G0	Gain Setting
0	0	0	1
0	0	1	2
0	1	0	4
0	1	1	8
1	0	0	16
1	0	1	32
1	1	0	64
1	1	1	128

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这里有个比较人性化的地方，无论是在哪个校准模式，只要完成了校准，就都会回到normal模式。因此，判断校准是否完成，就可以通过读MD01实现。然而，还有一种方法就是读DRDY低有效。关于这些ADI也有描述：

Calibration Sequences

The AD7705/AD7706 contain a number of calibration options, as previously outlined. Table 21 summarizes the calibration types, the operations involved, and the duration of the operations. There are two methods for determining the end of a calibration. The first is to monitor when DRDY returns low at the end of the sequence. This technique not only indicates when the sequence is complete, but also when the part has a valid new sample in its data register. This valid new sample is the result of a normal conversion that follows the calibration sequence. The second method for determining when calibration is complete is to monitor the MD1 and MD0 bits of the setup register. When these bits return to 0 following a calibration command, the calibration sequence is complete. This technique can indicate the completion of a calibration earlier than the first method can, but it cannot indicate when there is a valid new result in the data register. The time that it takes the mode bits, MD1 and MD0, to return to 0 represents the duration of the calibration. The sequence when DRDY goes low includes a normal conversion and a pipeline delay, t_p, to scale the results of this first conversion correctly. Note that t_p never exceeds 2000 × t_{CLKIN}. The time for both methods is shown in Table 21.

ysqjiangsu

3.时钟寄存器010

Table 18. Clock Register							
ZERO (0)	ZERO (0)	ZERO (0)	CLKDIS (0)	CLKDIV (0)	CLK (1)	FS1 (0)	FS0 (1)

Table 19. Clock Register Description	
Register	Description
ZERO	Zero. A zero must be written to these bits to ensure correct operation of the AD7705/AD7706. Failure to do so might result in unspecified operation of the device.
CLKDIS	Master Clock Disable Bit. Logic 1 in this bit disables the master clock, preventing it from appearing at the MCLK OUT pin. When disabled, the MCLK OUT pin is forced low. This feature allows the user the flexibility of either using the MCLK OUT as a clock source for other devices in the system, or turning off the MCLK OUT as a power-saving feature. When using an external master clock on the MCLK IN pin, the AD7705/AD7706 continue to have internal clocks and convert normally with the CLKDIS bit active. When using a crystal oscillator or ceramic resonator across Pin MCLK IN and Pin MCLK OUT, the AD7705/AD7706 clocks are stopped, and no conversions take place when the CLKDIS bit is active.
CLKDIV	Clock Divider Bit. With this bit at Logic 1, the clock frequency appearing at the MCLK IN pin is divided by 2 before being used internally by the AD7705/AD7706. For example, when this bit is set to Logic 1, the user can operate with a 4.9152 MHz crystal between Pin MCLK IN and Pin MCLK OUT, and internally the part operates with the specified 2.4576 MHz. With this bit at Logic 0, the clock frequency appearing at the MCLK IN pin is the frequency used internally by the part.
CLK	Clock Bit. This bit should be set in accordance with the operating frequency of the AD7705/AD7706. If the device has a master clock frequency of 2.4576 MHz (CLKDIV = 0) or 4.9152 MHz (CLKDIV = 1), this bit should be set to Logic 1. If the device has a master clock frequency of 1 MHz (CLKDIV = 0) or 2 MHz (CLKDIV = 1), this bit should be set to Logic 0. This bit sets up the appropriate scaling currents for a given operating frequency and, together with FS1 and FS0, chooses the output update rate for the device. If this bit is not set correctly for the master clock frequency of the device, the AD7705/AD7706 might not operate to specification.
FS1, FS0	Filter Selection Bits. Along with the CLK bit, FS1 and FS0 determine the output update rate, the filter's first notch, and the −3 dB frequency, as outlined in Table 20. The on-chip digital filter provides a sinc ³ (or (sinx/x) ³) filter response. In association with the gain selection, it also determines the output noise of the device. Changing the filter notch frequency, as well as the selected gain, impacts resolution. Table 5 through Table 8 show the effects of filter notch frequency and gain on the output noise and effective resolution of the part. The output data rate, or effective conversion time, for the device is equal to the frequency selected for the first notch of the filter. For example, if the first notch of the filter is selected at 50 Hz, a new word is available at a 50 Hz output rate, or every 20 ms. If the first notch is at 500 Hz, a new word is available every 2 ms. A calibration should be initiated when any of these bits are changed. The settling time of the filter to a full-scale step input is worst case 4 × 1/(output data rate). For example, with the filter-first notch at 50 Hz, the settling time of the filter to a full-scale step input is 80 ms maximum. If the first notch is at 500 Hz, the settling time is 8 ms maximum. This settling time can be reduced to 3 × 1/(output data rate) by synchronizing the step input change with a reset of the digital filter. In other words, if the step input takes place with the FSYNC bit high, the settling time is 3 × 1/(output data rate) from the time when the FSYNC bit returns low. The −3 dB frequency is determined by the programmed first notch frequency according to the relationship: $filter - 3\text{ dB frequency} = 0.262 \times filter - first\ notch\ frequency$

使用外部晶振4.9152MHz，显然要把CLKDIV设置成1的，否则会工作异常

Table 20. Output Update Rates				
CLK ¹	FS1	FS0	Output Update Rate	−3 dB Filter Cutoff
0	0	0	20 Hz	5.24 Hz
0	0	1	25 Hz	6.55 Hz
0	1	0	100 Hz	26.2 Hz
0	1	1	200 Hz	52.4 Hz
1	0	0	50 Hz	13.1 Hz
1	0	1	60 Hz	15.7 Hz
1	1	0	250 Hz	65.5 Hz
1	1	1	500 Hz	131 Hz

更新频率设置就靠这3bit了，显然500Hz要用1 1 1。然而，网上一大部分demo竟然这样误导：

```
1 /* 时钟寄存器bit定义 */
2 enum
3 {
4     CLKDIS_0    = 0x00, /* 时钟输出使能（当外接晶振时，必须使能才能振荡） */
5     CLKDIS_1    = 0x10, /* 时钟禁止（当外部提供时钟时，设置该位可以禁止MCK_OUT引脚输出时钟以省电 */
6
7     /*
8      * 2.4576MHz (CLKDIV=0 ) 或为 4.9152MHz (CLKDIV=1 ) , CLK 应置 “0”。
9      * 1MHz (CLKDIV=0 ) 或 2MHz (CLKDIV=1 ) , CLK 该位应置 “1”
10    */
11     CLK_4_9152M = 0x08,
12     CLK_2_4576M = 0x00,
13     CLK_1M      = 0x04,
14     CLK_2M      = 0x0C,
15
16     FS_50HZ     = 0x00,
17     FS_60HZ     = 0x01,
18     FS_250HZ    = 0x02,
19     FS_500HZ    = 0x04,
20     /*
21      * 当使用主时钟为 2.4576MHz 时，强烈建议将时钟寄存器设为 84H, 此时数据输出更新率为10Hz, 即每0.1S 输出一个新数据。
22      * 当使用主时钟为 1MHz 时，强烈建议将时钟寄存器设为80H, 此时数据输出更新率为4Hz, 即每0.25S 输出一个新数据
23    */
24     ZERO_0      = 0x00,
25     ZERO_1      = 0x80
26 };
```

注意是FS_500HZ = 0x04,看看怎样使用的：

```
1 | AD7705_WriteByte(CLKDIS_0 | CLK_4_9152M | FS_500HZ);
```

我们看一下宏定义
CLKDIS_0 = 0x00,CLK_4_9152M = 0x08,FS_500HZ = 0x04,逻辑或出来是0x0c,
这刷新频率是50Hz好吗？所以，正确的宏定义应该这样写：**FS_500HZ = 0x07**

4.数据寄存器011

5.测试寄存器100

4.5比较简单，就不说了

6.零点校正寄存器110

就是芯片内部自短接测试输出，感觉是输入失调电压的测试方法

7.全量程校正寄存器111

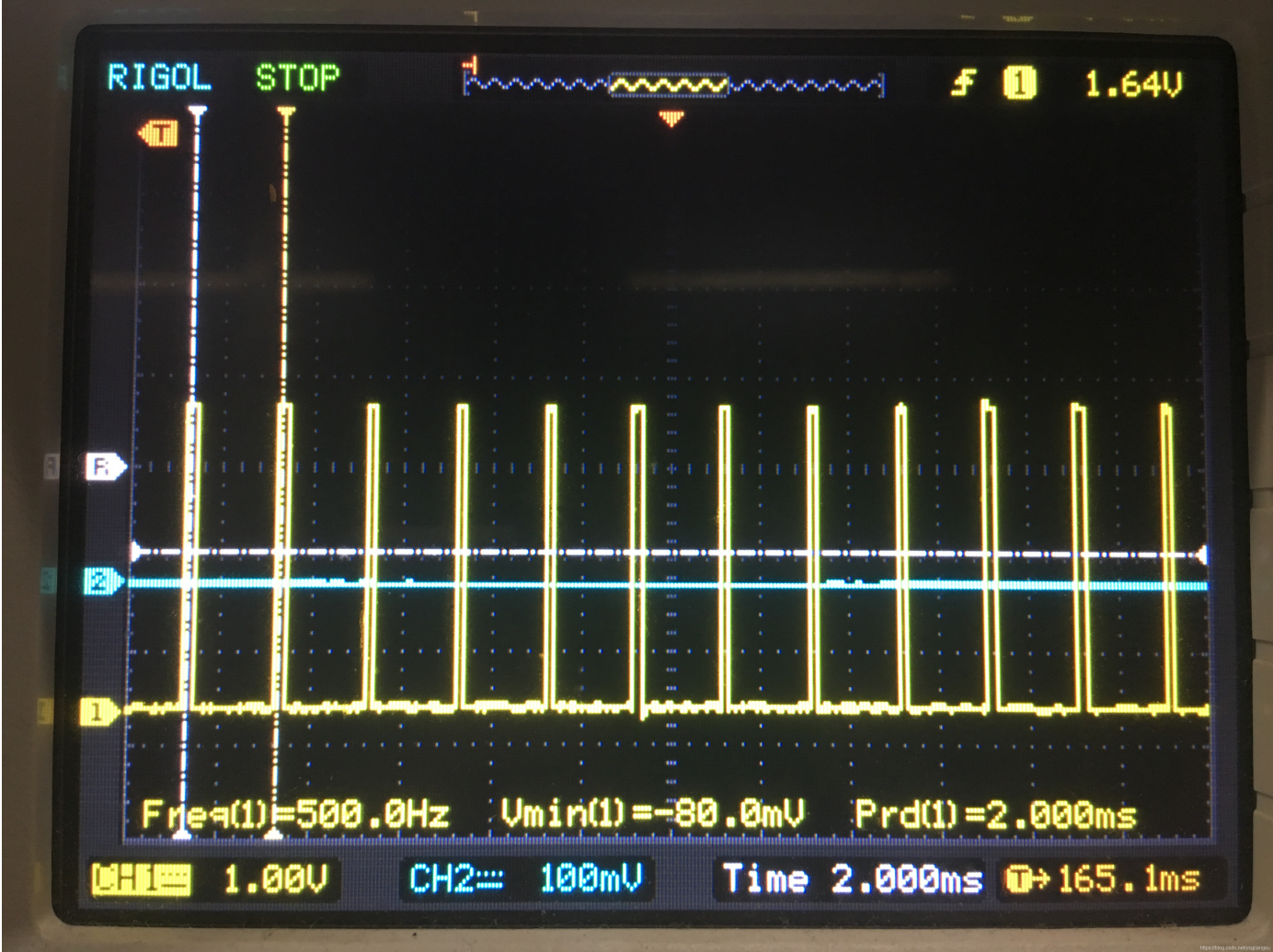
以单极性输入为例，量程是0-REF，那么芯片产生一个REF/GainSetting的电压，自测一下不就行了

Self-Calibration

A self-calibration is initiated on the AD7705/AD7706 by writing the appropriate values (0, 1) to the MD1 and MD0 bits of the setup register. In self-calibration mode with a unipolar input range, the zero-scale point used to determine the calibration coefficients is with the inputs of the differential pair internally shorted on the part (i.e., AIN(+) = AIN(-) = internal bias voltage on the AD7705, and AIN = COMMON = internal bias voltage on the AD7706). The PGA is set for the selected gain for this zero-scale calibration conversion, as per the G1 and G0 bits in the communication register. The full-scale calibration conversion is performed at the selected gain on an internally generated voltage of V_{REF}/selected gain.

使用看看

直接测试DRDY输出波形，如何？



500Hz的刷新频率出来了，为什么DRDY又被拉高了？因为如果两次更新之间，数据并未被读走，那么，赶紧在下次刷新之前提前拉高一下。提前多久？

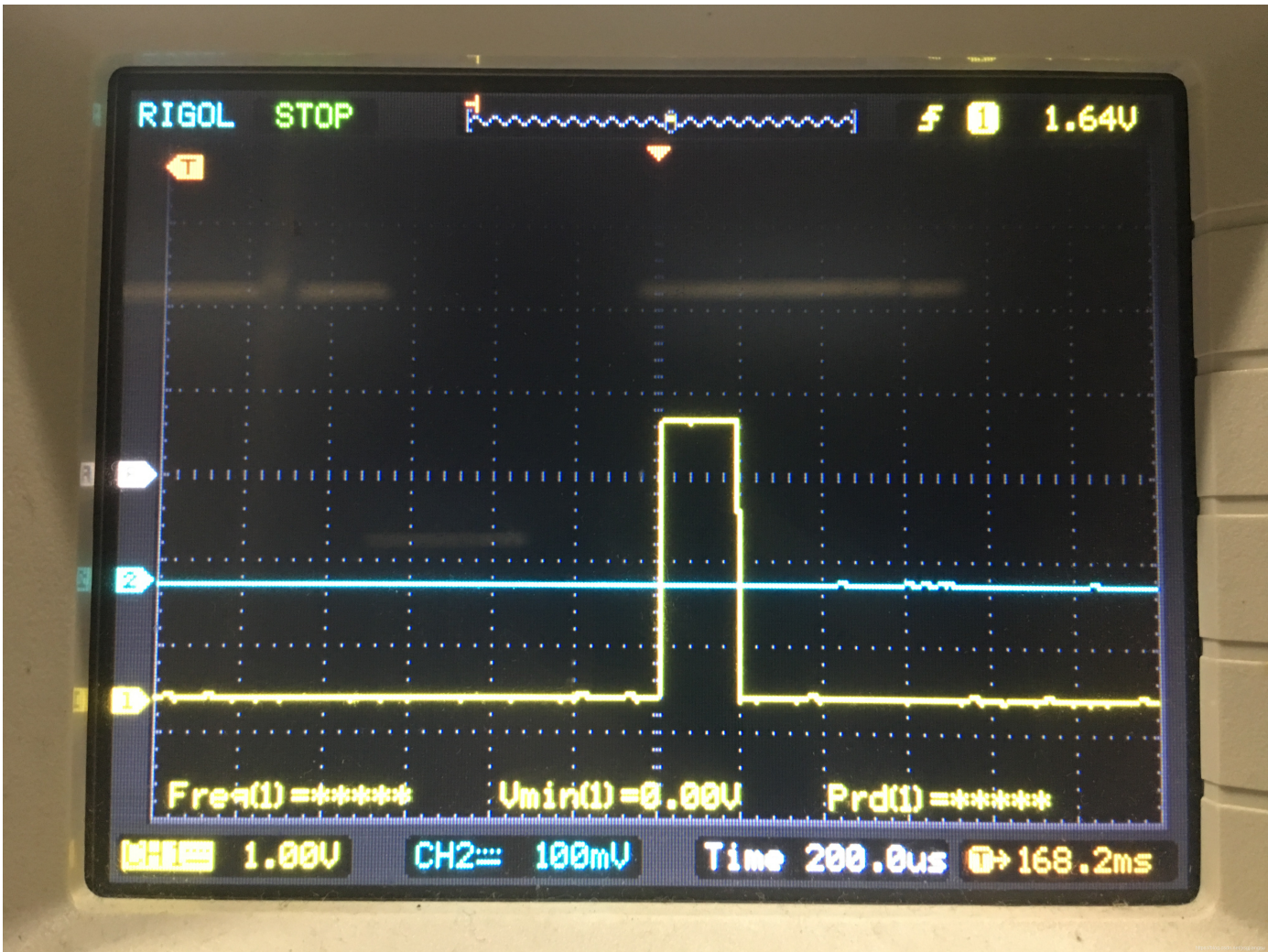
$500 \times F_{clk} = 500 * (1/2.4576MHz) = 203.45us$

可有依据？

12	DRDY	DRDY	Logic Output. A logic low on this output indicates that a new output word is available from the AD7705/AD7706 data register. The DRDY pin returns high upon completion of a read operation of a full output word. If no data read has taken place between output updates, the DRDY line returns high for 500 × t _{CLKIN} cycles prior to the next output update. While DRDY is high, a read operation should neither be attempted nor in progress to avoid reading from the data register as it is being updated. The DRDY line returns low after the update has taken place. DRDY is also used to indicate when the AD7705/AD7706 has completed its on-chip calibration sequence.
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我们的4.9152MHz分频后正是2.4576MHz，所以上式计算正确。

拉开一个高电平看看：



那500Hz刷新带来的问题是什么？

Table 7. Output RMS Noise vs. Gain and Output Update Rate @ 3 V									
Filter First Notch and O/P Data Rate	-3 dB Frequency	Typical Output RMS Noise in μ V							
		Gain of 1	Gain of 2	Gain of 4	Gain of 8	Gain of 16	Gain of 32	Gain of 64	Gain of 128
MCLK IN = 2.4576 MHz									
50 Hz	13.1 Hz	3.8	2.4	1.5	1.3	1.1	1.0	0.9	0.9
60 Hz	15.72 Hz	5.1	2.9	1.7	1.5	1.2	1.0	0.9	0.9
250 Hz	65.5 Hz	50	25	14	9.9	5.1	2.6	2.3	2.0
500 Hz	131 Hz	270	135	65	41	22	9.7	5.1	3.3
MCLK IN = 1 MHz									
20 Hz	5.24 Hz	3.8	2.4	1.5	1.3	1.1	1.0	0.9	0.9
25 Hz	6.55 Hz	5.1	2.9	1.7	1.5	1.2	1.0	0.9	0.9
100 Hz	26.2 Hz	50	25	14	9.9	5.1	2.6	2.3	2.0
200 Hz	52.4 Hz	270	135	65	41	22	9.7	5.1	3.3

Table 8. Peak-to-Peak Resolution vs. Gain and Output Update Rate @ 3 V									
Filter First Notch and O/P Data Rate	–3 dB Frequency	Typical Peak-to-Peak Resolution in Bits							
		Gain of 1	Gain of 2	Gain of 4	Gain of 8	Gain of 16	Gain of 32	Gain of 64	Gain of 128
MCLK IN = 2.4576 MHz									
50 Hz	13.1 Hz	16	16	15	15	14	13	13	12
60 Hz	15.72 Hz	16	16	15	14	14	13	13	12
250 Hz	65.5 Hz	13	13	13	13	12	12	11	11
500 Hz	131 Hz	10	10	10	10	10	10	10	10
MCLK IN = 1 MHz									
20 Hz	5.24 Hz	16	16	15	15	14	13	13	12
25 Hz	6.55 Hz	16	16	15	14	14	13	13	12
100 Hz	26.2 Hz	13	13	13	13	12	12	11	11
200 Hz	52.4 Hz	10	10	10	10	10	10	10	10

有效位直接干掉6bit，有点亏（就是最后6bit抖得像狗）。我倒是倾向于250Hz刷新，也就是4ms，仍然有13bit有效位，算是一种折中。